RANDOMIZED SMOOTHING AT HIGH CONFIDENCE: THE GAUSSIAN IS OPTIMAL FOR EVERY NOISE LAW

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Abstract. Randomized smoothing certifies a classifier by averaging a base classifier over a noise law q on $\mathbb{R}^d$. We consider certificates that hold for every base classifier and require only that the top class has probability at least $1-\alpha$ under the noise. We prove that if $\mathbb{E}\|\eta\|_2^2 \le d\sigma^2$ for $\eta \sim q$, then for $0 < \alpha < 1/4$ and $d \ge 1012\,\alpha^{-2}\log(16/\alpha)$ no such certificate exceeds $8\sigma\sqrt{\log(16/\alpha)}\,d^{1/p-1/2}$ in the ℓ_p norm, $2 \le p \le \infty$. The Gaussian $N(0,\sigma^2 I)$ certifies $\sigma\Phi^{-1}(1-\alpha)\,d^{1/p-1/2}$, so it is optimal up to an absolute constant factor at every confidence level. Earlier bounds for general noise laws allow the certified radius at fixed noise level to grow like $\alpha^{-1/2}$ as $\alpha \to 0$; the correct rate is $(\log 1/\alpha)^{1/2}$. The proof uses only the second moment of q: no product structure, symmetry, density or log-concavity.

1. Introduction

Let $f\colon \mathbb{R}^d \to \mathcal{Y}$ be a measurable classifier and let q be a Borel probability measure on $\mathbb{R}^d$. The smoothed classifier $g(x) = \arg\max_c \mathbb{P}_{\eta\sim q}(f(x+\eta)=c)$ is the object of randomized smoothing [4, 5, 2]. For the Gaussian $q = N(0,\sigma^2 I)$, Cohen, Rosenfeld and Kolter [2] showed by the Neyman–Pearson lemma that if the top class A has probability $p_A > 1/2$ at x, then $g(x+\delta) = A$ whenever $\|\delta\|_2 < \sigma\Phi^{-1}(p_A)$, and that this radius cannot be improved using p_A alone. Since the ℓ_∞ ball of radius r lies in the ℓ_2 ball of radius $r\sqrt{d}$, the same noise certifies the ℓ_∞ radius $\sigma\Phi^{-1}(p_A)/\sqrt{d}$.

Whether another noise law does better for ℓ_p, $p > 2$, has been studied from several directions. Blum, Dick, Manoj and Zhang [1] proved that a noise law which certifies ℓ_p radius ε for every base classifier with score gap δ must have $\mathbb{E}\eta_i^2 = \Omega(d^{1-2/p}\varepsilon^2(1-\delta)/\delta^2)$ for most coordinates i. Kumar, Levine, Goldstein and Feizi [3] showed that for a large class of product noise laws the certified ℓ_p radius decreases as $O(d^{1/p-1/2})$. Yang et al. [8] obtained a bound of the same order in d for a broad class of smoothing schemes that use only the probability of the top class, including schemes whose noise depends on the input, and Wu, Bojchevski, Kuvshinov and Günnemann [7] gave upper bounds of order $d^{-1/2}$ for the ℓ_2 radius under continuous symmetric noise.

Read at a fixed noise level σ per coordinate and confidence $p_A = 1-\alpha$, so that the score gap is $1-2\alpha$, the bound of [1] gives $\varepsilon = O(\sigma d^{1/p-1/2}\alpha^{-1/2})$, while the Gaussian certifies $\sigma\Phi^{-1}(1-\alpha)d^{1/p-1/2} \sim \sigma\sqrt{2\log(1/\alpha)}\,d^{1/p-1/2}$. The gap between $\alpha^{-1/2}$ and $(\log 1/\alpha)^{1/2}$ is the question of whether a non-Gaussian law can buy more radius at high confidence. The theorem below shows it cannot.

Definition 1.1. Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$ and $0 < \alpha < 1/2$. The *certified radius* $R_q(\alpha, \|\cdot\|)$ is the supremum of $\varepsilon \ge 0$ with the following property: for every measurable $f\colon \mathbb{R}^d \to \mathcal{Y}$, every $x \in \mathbb{R}^d$ and every class A with $\mathbb{P}_{\eta\sim q}(f(x+\eta)=A) \ge 1-\alpha$, one has $\mathbb{P}_{\eta\sim q}(f(x+\delta+\eta)=A) > 1/2$ for all $\|\delta\| \le \varepsilon$.

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The condition $\mathbb{P}(f(x+\delta+\eta)=A) > 1/2$ forces $g(x+\delta)=A$, so R_q is the largest radius that can be certified from the confidence level alone, for every base classifier. By the Cohen–Rosenfeld–Kolter bound, $R_{N(0,\sigma^2 I)}(\alpha, \|\cdot\|_2) = \sigma\Phi^{-1}(1-\alpha)$.

Theorem 1.2. *Let q be a Borel probability measure on $\mathbb{R}^d$ with $\mathbb{E}_{\eta\sim q}\|\eta\|_2^2 \le d\sigma^2$, where $\sigma > 0$. Let $0 < \alpha < 1/4$ and*

$$d \;\ge\; \frac{1012\,\log(16/\alpha)}{\alpha^2}.$$

Then for every $2 \le p \le \infty$,

$$R_q(\alpha, \|\cdot\|_p) \;\le\; 8\,\sigma\sqrt{\log(16/\alpha)}\; d^{1/p-1/2}.$$

Corollary 1.3. *Under the assumptions of Theorem 1.2, for every $2 \le p \le \infty$,*

$$R_q(\alpha, \|\cdot\|_p) \;\le\; 25\, R_{N(0,\sigma^2 I)}(\alpha, \|\cdot\|_p),$$

and the ratio of the two bounds tends to at most $8/\sqrt{2} < 5.7$ as $\alpha \to 0$.

The constants are not optimized. Only the second moment of q enters; the noise may be any law, with atoms, heavy tails or strong dependence between coordinates. The proof is an explicit construction: a base classifier that has probability at least $1-\alpha$ of the top class at x and probability at most $1/2$ at a point $x+v$ with $\|v\|_\infty = \varepsilon$. The shift v is a multiple of a sign vector, which is why the ℓ_p bounds follow from the ℓ_∞ one.

2. A criterion for failure

Lemma 2.1. *Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$. Suppose there are a Borel set $B_0 \subset \mathbb{R}^d$ and a vector v with $q(B_0) \ge 1/2$ and $q(v+B_0) \le \alpha$. Then $R_q(\alpha, \|\cdot\|) \le \|v\|$.*

Proof. Fix x and two classes $A \ne B$. Let $f(y) = B$ if $y \in x+v+B_0$ and $f(y) = A$ otherwise. Then $\mathbb{P}(f(x+\eta)=B) = q(v+B_0) \le \alpha$, so $\mathbb{P}(f(x+\eta)=A) \ge 1-\alpha$. At the point $x+v$, $\mathbb{P}(f(x+v+\eta)=B) = q(B_0) \ge 1/2$, so $\mathbb{P}(f(x+v+\eta)=A) \le 1/2$. Hence no $\varepsilon \ge \|v\|$ has the property of Definition 1.1. □

3. Proof of the theorem

Throughout, $\eta \sim q$, $B_r = \{y : \|y\|_2 \le r\}$, and $\varepsilon > 0$ is a number with $\varepsilon \le \sigma$ to be fixed at the end.

Lemma 3.1 (a thin shell)**.** *There is $\rho \in [2\sigma\sqrt{d},\, 3\sigma\sqrt{d} - \varepsilon\sqrt{d}\,]$ with $q(B_\rho) \ge 3/4$ and*

$$q\big(\rho < \|\eta\|_2 \le \rho + \varepsilon\sqrt{d}\big) \;\le\; \frac{2\varepsilon}{\sigma}.$$

Proof. By Markov's inequality $q(\|\eta\|_2 > 2\sigma\sqrt{d}) \le \mathbb{E}\|\eta\|_2^2/(4d\sigma^2) \le 1/4$, so $q(B_\rho) \ge 3/4$ for every $\rho \ge 2\sigma\sqrt{d}$. The interval $[2\sigma\sqrt{d}, 3\sigma\sqrt{d}]$ contains $N = \lfloor \sigma/\varepsilon \rfloor \ge 1$ disjoint half-open intervals $(\rho_k, \rho_k + \varepsilon\sqrt{d}]$, $\rho_k = 2\sigma\sqrt{d} + k\varepsilon\sqrt{d}$, $0 \le k < N$. The corresponding shells are disjoint, so one of them has mass at most $1/N \le 2\varepsilon/\sigma$, using $\lfloor t \rfloor \ge t/2$ for $t \ge 1$. □

Fix ρ as in Lemma 3.1 and let μ be the law of η conditioned on $\eta \in B_\rho$. For a sign vector $s \in \{-1,1\}^d$ write $X_s(y) = \langle y, s\rangle/\sqrt{d}$. Put

$$T = \sigma\sqrt{18\log 24}, \qquad T' = \sigma\sqrt{18\log(16/\alpha)}.$$

Lemma 3.2 (a direction)**.** *There is $s \in \{-1,1\}^d$ with $\mu(X_s > T) \le 1/3$ and $\mu(X_s < -T') \le \alpha/2$.*

Proof. Let s be uniform on $\{-1,1\}^d$, independent of $\eta \sim \mu$. For fixed $y \in B_\rho$, $\langle y, s\rangle$ is a sum of independent bounded terms $s_i y_i$, and Hoeffding's inequality gives, for $u > 0$,

$$\mathbb{P}_s\big(X_s(y) > u\big) \le \exp\Big(-\frac{u^2 d}{2\|y\|_2^2}\Big) \le \exp\Big(-\frac{u^2}{18\sigma^2}\Big),$$

since $\|y\|_2 \le \rho \le 3\sigma\sqrt{d}$; the same bound holds for $\mathbb{P}_s(X_s(y) < -u)$. Integrating over y, $\mathbb{E}_s\,\mu(X_s > T) \le 1/24$ and $\mathbb{E}_s\,\mu(X_s < -T') \le \alpha/16$. By Markov's inequality in s, $\mathbb{P}_s(\mu(X_s > T) > 1/3) \le 1/8$ and $\mathbb{P}_s(\mu(X_s < -T') > \alpha/2) \le 1/8$, so some s satisfies both bounds. □

Proof of Theorem 1.2. Take s from Lemma 3.2 and set

$$B_0 = \{y \in B_\rho :\ X_s(y) \le T\}, \qquad \varepsilon = \frac{T + T'}{\sqrt{d}}, \qquad v = -\varepsilon s.$$

First, $q(B_0) = q(B_\rho)\,\mu(X_s \le T) \ge \frac{3}{4} \cdot \frac{2}{3} = \frac{1}{2}$.

Next let $y = v + z$ with $z \in B_0$. Then $\|y\|_2 \le \rho + \|v\|_2 = \rho + \varepsilon\sqrt{d}$ and $X_s(y) = X_s(z) - \varepsilon\sqrt{d} \le T - (T + T') = -T'$. So y lies either in the shell $\{\rho < \|y\|_2 \le \rho + \varepsilon\sqrt{d}\}$ or in $B_\rho \cap \{X_s \le -T'\}$, and

$$q(v + B_0) \ \le\ \frac{2\varepsilon}{\sigma} + q(B_\rho)\,\mu(X_s \le -T') \ \le\ \frac{2\varepsilon}{\sigma} + \frac{\alpha}{2},$$

where $\mu(X_s \le -T') \le \alpha/2$ holds by the proof of Lemma 3.2, the Hoeffding bound being the same for the closed event.

It remains to check $2\varepsilon/\sigma \le \alpha/2$ and $\varepsilon \le \sigma$. For $\alpha < 1/4$ we have $\log(16/\alpha) > \log 64$, hence

$$\frac{T + T'}{\sigma} = \sqrt{18}\Big(\sqrt{\log 24} + \sqrt{\log(16/\alpha)}\Big) \le \sqrt{18}\Big(\sqrt{\tfrac{\log 24}{\log 64}} + 1\Big)\sqrt{\log(16/\alpha)} \le 7.952\sqrt{\log(16/\alpha)}.$$

The condition $\varepsilon \le \alpha\sigma/4$ reads $\sqrt{d} \ge 4(T+T')/(\alpha\sigma)$, which holds when $d \ge 16 \cdot 7.952^2\,\alpha^{-2}\log(16/\alpha)$, and $16 \cdot 7.952^2 < 1012$. Then also $\varepsilon \le \sigma$. So $q(v + B_0) \le \alpha$, and Lemma 2.1 gives, since v is ε times a sign vector,

$$R_q(\alpha, \|\cdot\|_p) \le \|v\|_p = \varepsilon\, d^{1/p} = \frac{T + T'}{\sigma}\,\sigma\, d^{1/p - 1/2} \le 8\,\sigma\sqrt{\log(16/\alpha)}\, d^{1/p - 1/2}.$$ □

Proof of Corollary 1.3. For $p \ge 2$, $\|\delta\|_2 \le d^{1/2 - 1/p}\|\delta\|_p$, so the Gaussian certifies the ℓ_p radius $\sigma\Phi^{-1}(1-\alpha)\, d^{1/p - 1/2}$. The function $\alpha \mapsto 8\sqrt{\log(16/\alpha)}/\Phi^{-1}(1-\alpha)$ is at most 24.19 on $(0, 1/4)$, with the largest values near $\alpha = 1/4$, and tends to $8/\sqrt{2}$ as $\alpha \to 0$ because $\Phi^{-1}(1-\alpha) \sim \sqrt{2\log(1/\alpha)}$. □

4. Remarks

Remark 4.1. The Gaussian value $\sigma\Phi^{-1}(1-\alpha)/\sqrt{d}$ in ℓ_∞ is exact for $N(0, \sigma^2 I)$: the shift $v = \varepsilon(1, \dots, 1)$ and the half-space classifier $f(y) = B$ on $\{\langle y - x, (1, \dots, 1)\rangle \ge \sigma\sqrt{d}\,\Phi^{-1}(1-\alpha)\}$ attain it. Theorem 1.2 says that no other law with the same average per-coordinate second moment improves this by more than a constant factor, at any confidence level.

Remark 4.2. The noise level is measured by the average second moment $d^{-1}\mathbb{E}\|\eta\|_2^2$, as in [3]; the bound of [1] is stated for most individual coordinates.

Remark 4.3. The condition $d \ge C\alpha^{-2}\log(1/\alpha)$ comes from the shell step, which needs a shell of relative mass $O(\alpha)$, and is not optimized. Three questions remain: whether the Gaussian is exactly optimal among laws with a given second moment; whether the condition on d can be removed; and what happens for input-dependent smoothing, where the noise law varies with x [6]. The argument above uses a single law q at x and at $x + v$ and does not apply directly to the last case.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

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